selective prediction for ECG diagnostic traig

Abstract

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Deep neural networks are typically trained to minimize an empirical loss function. While they often achieve high accuracy, they are notoriously overconfident. A model might assign a near-certain probability to a classification even when it is seeing an image or a piece of data far outside its training distribution. Standard techniques for uncertainty quantification, such as Bayesian Neural Networks or Monte Carlo Dropout, often require significant computational overhead, modifications to the training architecture, or assumptions about the underlying data distribution that may not hold in practice.

Conformal prediction works by shifting the focus from finding the exact output to identifying a set of outputs that contains the true value with a high degree of probability. The process begins with a calibration set, which is a small, held-out portion of data that the model has not seen during training. This set is used to calibrate the model's error distribution. Within this framework, researchers define a non-conformity score, which acts as a measure of how strange or unexpected a new data point is compared to the calibration data. For a regression problem, this is often calculated as the absolute residual between the true value and the model's prediction.

By calculating these scores for the entire calibration set, one can determine a specific threshold at a desired significance level. If an analyst desires a certain percentage of confidence, they choose a corresponding significance level and identify the quantile of the non-conformity scores. For any new, unseen input, this threshold is used to construct a prediction interval. Because of the mathematical properties of exchangeability within the data, this interval is guaranteed to contain the true label at least the specified percentage of the time, regardless of the distribution of the data or the internal complexities of the neural network.

The primary advantages of this approach include distribution-free coverage, model agnosticism, and computational efficiency. Unlike Bayesian methods, this framework does not assume that the data follows a specific distribution, such as a Gaussian distribution. As long as the data is exchangeable, the coverage guarantee holds. Furthermore, users can apply conformal methods to any architecture, such as Transformers or convolutional networks, without needing to retrain the underlying model. The only requirement is a single pass over the small calibration set, which avoids the need for costly re-training or complex sampling procedures.

Practical applications for this technology are vast. In healthcare, a model can provide a set of potential diagnoses rather than just one. If the set contains only one, the doctor can be confident; if the set is large, the system signals that the model is uncertain and requires manual expert review. In autonomous systems, instead of predicting a single trajectory for a pedestrian, the system generates a safety zone or a set of potential paths, ensuring the vehicle remains safe even if the prediction is not

exact. In finance, conformal regression is used to provide valid price intervals for assets, giving risk managers a mathematically grounded scenario for potential fluctuations.

While powerful, this methodology is not a panacea. The size of the prediction set is inversely related to the accuracy of the underlying model. If a deep learning model is poor, the conformal interval will be very wide, potentially becoming uselessly uninformative. The goal of conformalizing a model is essentially to ensure that if a model does not know the answer, it communicates that through a large, honest uncertainty range rather than a misleadingly precise, incorrect answer. Future research is moving toward adaptive prediction sets, where the size of the set depends on the difficulty of the input, and conditional coverage, which aims to ensure that the error rates are consistent across different subpopulations of data. By bridging the gap between raw predictive power and statistical rigor, this field is helping to move artificial intelligence systems into real-world applications where reliability is a fundamental requirement.

Keywords:

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1. Introduction

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2. Related Work

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3. Proposed Evidential Framework

Table 1
Validation Set Class Counts and Sample Sufficiency Split by Clinical Tier (exact threshold $n \geq \lceil 1/\alpha \rceil - 1$, tier-specific)

Tier	Target α	Supported	Unsupported
Critical	0.02	AMI(306), IMI(326), AFIB(151), _AVB(83)	PMI(2)[a], AFLT(7), LMI(20)[b]
Important	0.05	ISCA(92), ISC_(125), ISCI(39), CLBBB(54)	WPW(7), PSVT(3), SVTAC(3), TRIGU(2), BIGU(8), SVARR(15)[b]
Benign	0.10	STTC(225), LVH(210), IRBBB(112), NST_(75), IVCD(78), LAFB/LPFB(181), CRBBB(55), LAO/LAE(43), SARRH(77), SBRAD(64), STACH(83), PACE(29), RVH(12), RAO/RAE(10)[c]	SEHYP(3), ILBBB(7)
Excluded	n/a	NORM(955), SR(1670)	—

[a] PMI ($n = 2$): best achievable guarantee is $\alpha_{\min} = 1/(n+1) \approx 0.33$, looser than even the Benign target. No tier target in this scheme is mathematically achievable; excluded from per-class FNR reporting unless calibration data is pooled across datasets (Section [X]).
[b] Exceeds the originally-used flat $n \geq 10$ rule but falls short of the exact tier-specific floor required to achieve its stated target α.
[c] Meets the exact floor by a margin of exactly one calibration example; reported with caution, not treated as fully stable.

4. Methodology

4.1. Dataset and Evaluation Splits

We use PTB-XL [1], a multi-label 12-lead ECG dataset distributed with a patient-stratified `strat_fold` column (values 1–10): no patient's records appear in more than one fold, and class distributions are held approximately consistent across folds. We adopt the train/validation/test convention established in the original PTB-XL benchmarking work [2] and apply it throughout this study: folds 1–8 for training (17,418 records), fold 9 for validation (2,183 records), and fold 10 for test (2,198 records).

Fold 9 serves two distinct roles in this work, and we make this explicit because it directly determines the sample-sufficiency results in Section 4.3.3. During model training, fold 9 is the validation set used for checkpoint selection and early stopping, in the conventional sense. For the acuity-weighted risk control procedure (Section 4.3), the same 2,183 records additionally serve as the calibration set used to fit each class's conformal threshold. These are not two independent samples but a single physical fold performing both functions, and a class's representation during training can differ substantially from its representation during calibration as a result: AFLT, for example, has 59 positive examples across the training folds (1–8) but only 7 in fold 9, and it is this latter count, not the training-time count, that constrains its calibration in Table 1.

Fold 10 is untouched by both training and calibration; it is reserved exclusively for final evaluation, so that the risk-coverage results and achieved-false-negative-rate-versus-target comparisons reported in Section [evaluation section reference] are measured on data the pipeline has not seen in any capacity.

One consequence of reusing the validation fold as the calibration set is worth flagging directly: because checkpoint selection during training is itself informed by performance on fold 9, the nonconformity scores computed on this fold for calibration are not entirely independent of the model-selection process that produced f_θ. This is standard practice in conformal pipelines built on top of a model selected via the same held-out fold, but it is a narrower notion of exchangeability than a fully independent three-way split would provide, and we report it as a stated limitation of the experimental design rather than treat the calibration set as wholly unrelated to model development.

4.2. Problem Setup and Notation

4.3. Acuity-Weighted Class-Conditional Risk Control

4.3.1. Class granularity

We instantiate acuity-weighted risk control at the subdiagnostic (23-class) and rhythm (12-class) task granularities, rather than the coarser 5-class superdiagnostic task or the full 71-statement level. The superdiagnostic level conflates conditions of substantially different clinical urgency within a single class (e.g., the CD superclass contains both complete AV block and isolated incomplete bundle branch block); the full statement level, while more granular, produces calibration sets too sparse for several rare but high-acuity statements to support a stable per-class quantile estimate. The subdiagnostic and rhythm granularities provide classes that are both clinically distinguishable by acuity and adequately represented in the PTB-XL calibration folds.

4.3.2. Acuity tier definition

Each class is assigned to one of three ordinal acuity tiers—Critical, Important, Benign—together with a fourth, Excluded, category for baseline/negative findings (Table 1). We do not derive this tiering from a single canonical clinical risk-stratification instrument; rather, following the precedent established by recent conformal selective-prediction work in clinical triage [3], we define the tiering directly through manual clinical-judgment grouping of SCP-ECG classes, and treat it as a stated, documented modeling assumption rather than a claim of ground truth. The tier boundaries are broadly consistent with how the AHA/ACC/HRS practice standards for electrocardiographic monitoring [4] and the AHA/ACCF/HRS recommendations for ECG standardization, Part VI [5], stratify analogous conditions by monitoring priority and ischemic risk, though no formal one-to-one derivation from those instruments is performed. Table 1 documents, for every grouping, the rationale underlying its tier assignment; two groupings in particular reflect a conservative rather than a precise classification—MI subtypes, which cannot be separated into acute and chronic presentations at this granularity, and the _AVB bucket, which cannot be disentangled from high-grade block—and both are treated as Critical by default rather than assumed benign, as stated explicitly in the table.

4.3.3. Calibration procedure

Given the tier assignment, each tier is mapped to a target per-class miscoverage rate α_{tier} (Table 1), and class-conditional split conformal calibration is performed independently per class following Algorithm 1 (lines 4–13): for each class k, a finite-sample-corrected empirical quantile q_k of the nonconformity scores $\{1 - f_\theta(x_i)_k\}$ is computed over the calibration examples positive for class k, and the resulting per-class thresholds define the multi-label prediction set $\mathcal{C}(x) = \{k \in \mathcal{Y} : 1 - f_\theta(x)_k \leq q_k\}$.

This procedure can only deliver the stated target α_k if the calibration set contains sufficient positive examples for class k. With n_k such examples, the tightest miscoverage rate the quantile step in Algorithm 1 can guarantee is $\alpha_{\min} = 1/(n_k + 1)$; the target α_k is therefore attainable only if $n_k \geq \lceil 1/\alpha_k \rceil - 1$. Because this bound scales with the target itself, we apply it per tier rather than as a single fixed cutoff: $n_k \geq 49$ for the Critical tier, $n_k \geq 19$ for the Important tier, and $n_k \geq 9$ for the Benign tier. Table 1 reports the positive-example count for every class against its tier-specific requirement. The table floor is a necessary condition for the quantile to exist, not a sufficient condition for the binomial LTT to pass at $\delta/24$. With Bonferroni correction across 24 classes, the actual minimum n at $\alpha = 0.02$ is approximately 272. AFIB(151) and _AVB(83) both exceed 49 but fall below 272.

Algorithm 1: Acuity-Weighted Class-Conditional Risk Control (AW-CCRC)

Input: f_θ (classifier); $\mathcal{D}_{\text{cal}}=\{(x_i,y_i)\}_{i=1}^m$ ($y_i\subseteq\mathcal{Y}$, multi-label); $\mathcal{Y}$, $|\mathcal{Y}|=K=35$;
$T{:}\mathcal{Y}\to\{\text{Crit},\text{Imp},\text{Ben}\}$ (Alg. 2); $\mathcal{A}=\{\alpha^{(1)},\ldots,\alpha^{(S)}\}$; $\delta=0.10$; $\Lambda\subset[0,1]$, $|\Lambda|=G=1000$

Output: $\{\lambda_c^{(k,s)}\}$: calibrated thresholds ($\varnothing$ if uncertifiable)
$\{\widehat{\text{FNR}}_k^{(s)}\}$: empirical miss rate on test fold
Δ: cross-configuration stability report

`// Preprocessing - config-independent`
foreach $k\in\mathcal{Y}$ **do** $\text{tier}_k\leftarrow T(k)$
$\mathcal{Y}_{\text{sup}}\leftarrow\{k\in\mathcal{Y}:n_k^+\geq\lceil 1/\alpha_k\rceil-1\}$ `// min. samples for quantile to exist`
$\delta_k\leftarrow\delta\,/\,|\mathcal{Y}_{\text{sup}}|$ `// Bonferroni: simultaneous guarantee over` $|\mathcal{Y}_{\text{sup}}|=24$ `supported classes`

`// Per-configuration calibration and evaluation`
for $s\leftarrow 1$ **to** S **do**
  for *each* $k\in\mathcal{Y}$ **do** `// calibrate each class`
    $\alpha_k\leftarrow\alpha^{(s)}(\text{tier}_k)$; $\mathcal{S}_k\leftarrow\{f_\theta(x_i)_k:k\in y_i\}$; $n_k\leftarrow|\mathcal{S}_k|$
    if $(1-\alpha_k)^{n_k}>\delta_k$ **then** `// uncertifiable:` n_k `too small to reject` H_0 `at zero misses`
      $\lambda_c^{(k,s)}\leftarrow\varnothing$
    else
      $\lambda_c^{(k,s)}\leftarrow 0$ `// default: flag all (trivial,` $\widehat{\text{FNR}}=0$`)`
      for $\lambda_g\in\Lambda$ *in* decreasing *order* **do** `// largest` λ `rejecting` H_0
        $k_g\leftarrow\big|\{s'\in\mathcal{S}_k:s'<\lambda_g\}\big|$ `// miss count; non-decreasing in` λ_g
        $p_g\leftarrow\Pr\big[X\leq k_g\mid X\sim\text{Bin}(n_k,\alpha_k)\big]$ `// one-sided p-value:` $H_0{:}\,\text{FNR}(\lambda_g)\geq\alpha_k$
        if $p_g\leq\delta_k$ **then** $\lambda_c^{(k,s)}\leftarrow\lambda_g$; **break**
      end
    end
  end
  foreach *test record* x_j **do** $\mathcal{C}^{(s)}(x_j)\leftarrow\{k:\lambda_c^{(k,s)}\neq\varnothing,\ f_\theta(x_j)_k\geq\lambda_c^{(k,s)}\}$ `// predicted positives`
  foreach *certified* k **do** $\widehat{\text{FNR}}_k^{(s)}\leftarrow\hat{\Pr}\big[k\in y_j\wedge k\notin\mathcal{C}^{(s)}(x_j)\mid k\in y_j\big]$ `// empirical test miss rate`
end

$\Delta\leftarrow\text{Compare}\big(\{\widehat{\text{FNR}}_k^{(s)}\}_{s=1}^S,\ \text{RiskCoverageCurve}^{(s)},\ \text{FlagRate}^{(s)}\big)$
return $\{\lambda_c^{(k,s)}\}$, $\{\widehat{\text{FNR}}_k^{(s)}\}$, Δ

Three classes exceed a naive fixed threshold of $n \geq 10$ yet fall below the exact requirement for their assigned tier: LMI ($n = 20$ against a requirement of 49) and SVARR ($n = 15$ against 19). One Benign-tier class, RAO/RAE ($n = 10$), meets its requirement ($n \geq 9$) by a margin of a single example. These classes remain in the calibration procedure, but their resulting thresholds are reported in subsequent tables with an explicit finite-sample caveat rather than presented as equivalent in reliability to classes with substantially larger calibration support.

PMI is excluded from calibration entirely. With only $n = 2$ positive calibration examples, the best guarantee achievable under Algorithm 1 is $\alpha_{\min} \approx 0.33$, looser than every tier target in this scheme, including the most permissive (Benign, $\alpha = 0.10$). No re-assignment of PMI's target α within the tier structure resolves this, since the limitation is one of available sample size rather than tier placement. We therefore omit PMI from class-conditional calibration and from per-class false-negative-rate reporting in Section [evaluation section reference], and report this omission explicitly rather than present a threshold the calibration data cannot support.

4.3.4. Sensitivity analysis

Because the numeric α values attached to each tier are a modeling choice rather than an externally validated quantity, we repeat the full calibration and evaluation procedure under two additional configurations — a stricter mapping ($\alpha = 0.01/0.03/0.08$ for Critical / Important / Benign) and a looser one ($\alpha = 0.05/0.08/0.15$) — alongside the primary configuration (0.02/0.05/0.10). We report per-class false-negative-rate control, the risk–coverage ordering across methods, and the operational auto-confirm rate for all three configurations side by side (Section [evaluation section ref]). Findings that hold across all three configurations are reported as primary results; any metric sensitive to the specific tier-to-α mapping is reported explicitly as such, rather than presented only under the configuration most favorable to the method.

5. Experimental Results

5.1. Experimental Setup

5.2. Acuity-Weighted Class-Conditional Risk Control: Results

5.2.1. Validity of the testing procedure

Certification used the Learn-Then-Test (LTT) framework with a one-sided binomial risk test and a monotone-boundary search over the threshold λ, controlling the family-wise error rate at $\delta = 0.10$ across the 24 supported classes via Bonferroni correction (per-class $\delta \approx 0.0042$). The validity of this configuration was assessed by Monte Carlo simulation ($n_{\text{trials}} = 1000$, $n_{\text{cal}} = 2000$, target $\alpha = 0.10$, positive and negative nonconformity scores drawn from Beta $(7, 3)$ and Beta $(3, 7)$ respectively, with the true false-negative rate (FNR) evaluated analytically from the Beta CDF). The adopted binomial monotone-boundary variant produced an empirical violation rate of 0.078, within the permitted bound $\delta = 0.10$, at a mean certified threshold $\bar{\lambda}_c = 0.497$ and mean FNR 0.088 (Table 6). Two alternative estimators were evaluated for reference: a Hoeffding bound with grid-Bonferroni correction (violation = 0.000, $\bar{\lambda}_c = 0.418$, mean FNR 0.033) and a Hoeffding bound with monotone boundary (violation = 0.000, $\bar{\lambda}_c = 0.473$, mean FNR 0.067). All three configurations satisfied the coverage requirement.

5.2.2. Primary calibration outcomes

Across the 24 supported classes, the acuity-weighted procedure certified a class-conditional FNR guarantee at the tier-specific target for 11 classes, returned a trivial threshold ($\lambda = 0$, indicating no discriminative selectivity) for 2 classes, and returned no certificate for 11 classes (Table 4). For every certified class, the FNR measured on the held-out test fold (fold 10) fell at or below the class's target

α. In the Critical tier ($\alpha = 0.02$), AMI and IMI were certified, with test FNRs of 0.0131 and 0.0061 at certified thresholds $\lambda_c = 0.0060$ and 0.0020 and corresponding flag rates of 54.0% and 76.1%. The remaining three Critical-tier classes (AFIB, LMI, _AVB) were uncertifiable. In the Important tier ($\alpha = 0.05$), no class was certified: four classes were uncertifiable and ISC_ returned a trivial threshold. In the Benign tier ($\alpha = 0.10$), 9 of 14 classes were certified, with test FNRs ranging from 0.0000 (NST_) to 0.0901 (STTC) and flag rates ranging from 9.8% (STACH) to 96.1% (IVCD).

Two of the uncertifiable Critical/Important results were associated with calibration-fold sample size rather than with test FNR. AFIB ($n^+_{\text{val}} = 151$) did not meet the positive-count requirement for certification at $\alpha = 0.02$, and _AVB ($n^+_{\text{val}} = 83$) did not meet the requirement at $\alpha = 0.02$.

5.2.3. Comparison against uncorrected calibration baselines

To assess whether per-class miscoverage control requires the statistical correction applied by the binomial LTT procedure, two baselines without such correction were evaluated: temperature-scaled thresholding [**?**] and MC Dropout thresholding [**?**]. Each baseline selects, per class, the threshold λ at which validation-fold FNR first falls at or below α_{tier}, with no adjustment for sampling variability between the validation and test folds.

Across the 24 supported classes, the binomial LTT procedure produced no violations of the target α_{tier} on the test fold (0/24, 0%). Both uncorrected baselines violated their target on 14 of 24 classes (58%) (Table 2).

Table 2
Violation rate against target α_{tier}, test fold, by calibration method

Method	**Guarantee type**	**Violations**	**Rate**
LTT binomial	Statistical (binomial LTT)	0/24	0%
Temperature scaling (uncorrected)	Empirical	14/24	58%
MC Dropout (uncorrected)	Empirical	14/24	58%

These 14 violations are not concentrated among the classes LTT fails to certify. Of the 11 classes certified by LTT, 8 (AMI, IMI, IVCD, LVH, NST_, SARRH, SBRAD, STTC) violate α_{tier} under both uncorrected baselines on the test fold; the remaining 3 certified classes — IRBBB, LAFB/LPFB, STACH — do not violate under either baseline on this fold. Conversely, of the 13 classes LTT cannot certify (uncertifiable or trivial), 7 (_AVB, CLBBB, ISCA, ISCI, ISC_, RAO/RAE, CRBBB) do not violate α_{tier} under the uncorrected baselines on this fold, while the remaining 6 (AFIB, LMI, SVARR, LAO/LAE, PACE, RVH) do. The uncorrected baselines' agreement with the target on these 7 classes is an outcome of this particular validation/test split rather than a property guaranteed to hold under resampling, since the baseline procedure carries no statistical correction for sampling variability between folds. One of the seven, RAO/RAE, holds at exact equality (test FNR $= \alpha = 0.10$), with no margin between the observed value and the target.

Table 3 reports threshold and test-fold FNR for the 8 classes on which LTT certifies and both uncorrected baselines violate. In all 8 rows, the uncorrected threshold exceeds the LTT threshold for both baselines ($\lambda_{\text{TS}} > \lambda_{\text{LTT}}$ and $\lambda_{\text{MC}} > \lambda_{\text{LTT}}$, with no exception across the 16 comparisons). A higher threshold corresponds to a lower flag rate and, by the definition of FNR used here, a larger fraction of true positives falling below threshold and being scored negative — the numeric pattern underlying the violations in Table 2. The LTT threshold in each row is set by a procedure that incorporates the sampling gap between the validation fold, on which λ is chosen, and the test fold, on which FNR is subsequently measured; the uncorrected procedure selects the threshold at which validation-fold FNR first meets α_{tier}, with no explicit allowance for that gap.

Table 3
Threshold (λ) and test-fold FNR for the 8 classes where LTT certifies and both uncorrected baselines violate α_{tier}

Class	Tier	α	LTT		TS		MC	
			λ	FNR	λ	FNR	λ	FNR
AMI	Critical	0.02	0.0060	0.0131	0.0287	0.0261	0.0254	0.0261
IMI	Critical	0.02	0.0020	0.0061	0.0424	0.0245	0.0347	0.0214
IVCD	Benign	0.10	0.0010	0.0127	0.0088	0.1392	0.0065	0.1392
LVH	Benign	0.10	0.0260	0.0514	0.0954	0.1121	0.0812	0.1028
NST_	Benign	0.10	0.0010	0.0000	0.0161	0.1169	0.0121	0.1169
SARRH	Benign	0.10	0.0070	0.0260	0.0293	0.1299	0.0245	0.1299
SBRAD	Benign	0.10	0.0010	0.0156	0.0417	0.1250	0.0298	0.1250
STTC	Benign	0.10	0.0310	0.0901	0.0629	0.1171	0.0491	0.1171

Temperature scaling was also evaluated for its effect on aggregate calibration quality, independent of the per-class violation analysis above. Fitting $T = 1.115$ reduced expected calibration error (ECE) from 0.0049 to 0.0023, a 53% relative reduction. This improvement in aggregate calibration is distinct from per-class adherence to a false-negative-rate target: temperature scaling achieves a substantially lower ECE while simultaneously violating α_{tier} on 14 of 24 classes (Table 2), indicating that improved aggregate calibration does not by itself imply per-class control of the false-negative rate.

5.2.4. Comparison of tiered and uniform-α strategies

The tiered assignment was compared against three uniform-α strategies ($\alpha = 0.02$, 0.05, 0.10 applied identically to all classes) (Figure 1; Table 5). Relative to the tiered assignment, the uniform $\alpha = 0.02$ strategy failed to certify 9 Benign-tier classes that the tiered assignment certified (efficiency losses). The uniform $\alpha = 0.10$ strategy certified 6 Critical- and Important-tier classes at a target looser than their assigned tier required (safety failures); for these classes the published guarantee exceeded the tier's α (e.g., AFIB certified at an effective FNR bound of 0.054 against a Critical-tier requirement of 0.02). The two effect sets were disjoint (intersection = 0): no class appeared in both the efficiency-loss and safety-failure sets. A further 9 classes were unaffected by the choice of strategy, being uncertifiable or trivial under all configurations. In total, 15 of 24 supported classes (63%) had a different certification outcome under at least one uniform strategy than under the tiered assignment.

5.2.5. Discrimination and operating points

Per-class discrimination, summarised by the area under the receiver operating characteristic curve (AUROC), ranged from 0.746 to 0.998 across the supported classes (Figure 2). Certification outcome and AUROC were not aligned in rank: several high-AUROC classes were uncertifiable (AFIB, AUROC = 0.983; CLBBB, AUROC = 0.997), while a lower-AUROC class was certified (IVCD, AUROC = 0.746). The certified configurations yielded 34 risk-coverage operating points across 16 classes and the three α settings (Figure 3), with flag rate and test FNR varying by tier and target as tabulated in Table 4.

6. Discussion

6.1. Acuity-Weighted Class-Conditional Risk Control: Discussion

The central hypothesis was that a single uniform miscoverage target cannot simultaneously satisfy the differing reliability requirements of high- and low-acuity diagnostic classes, and that tier-specific

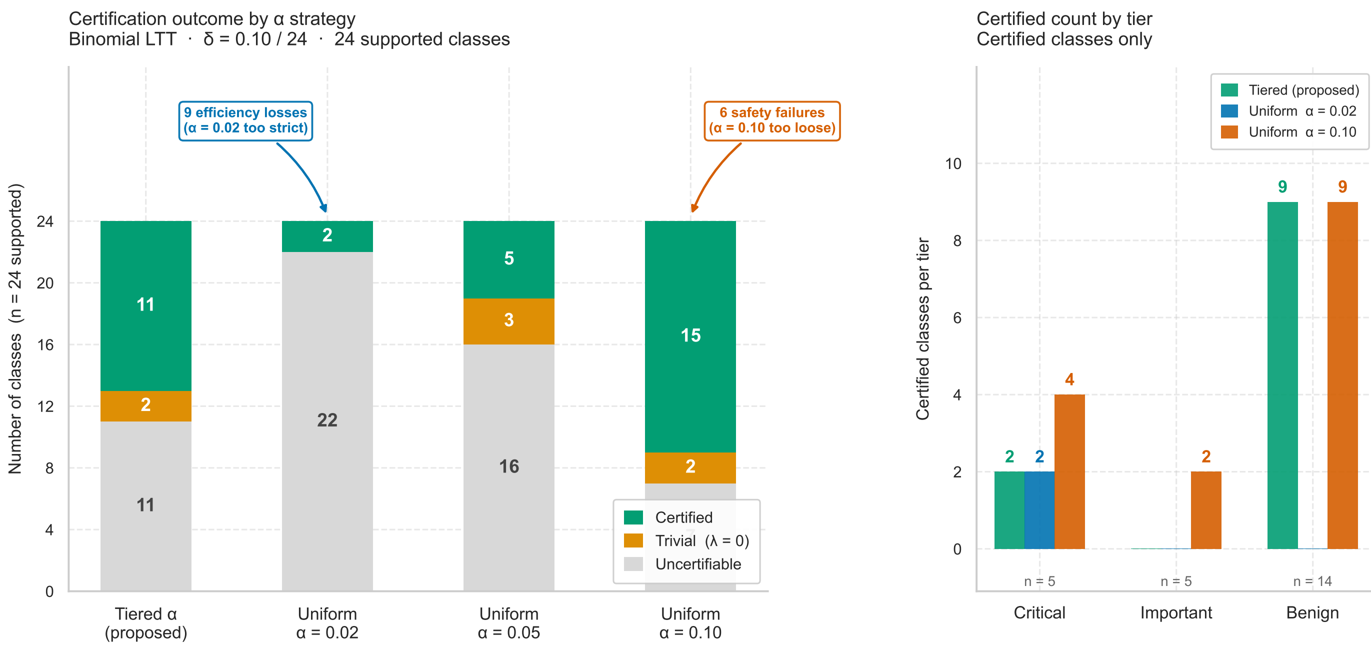

Figure 1: Caption

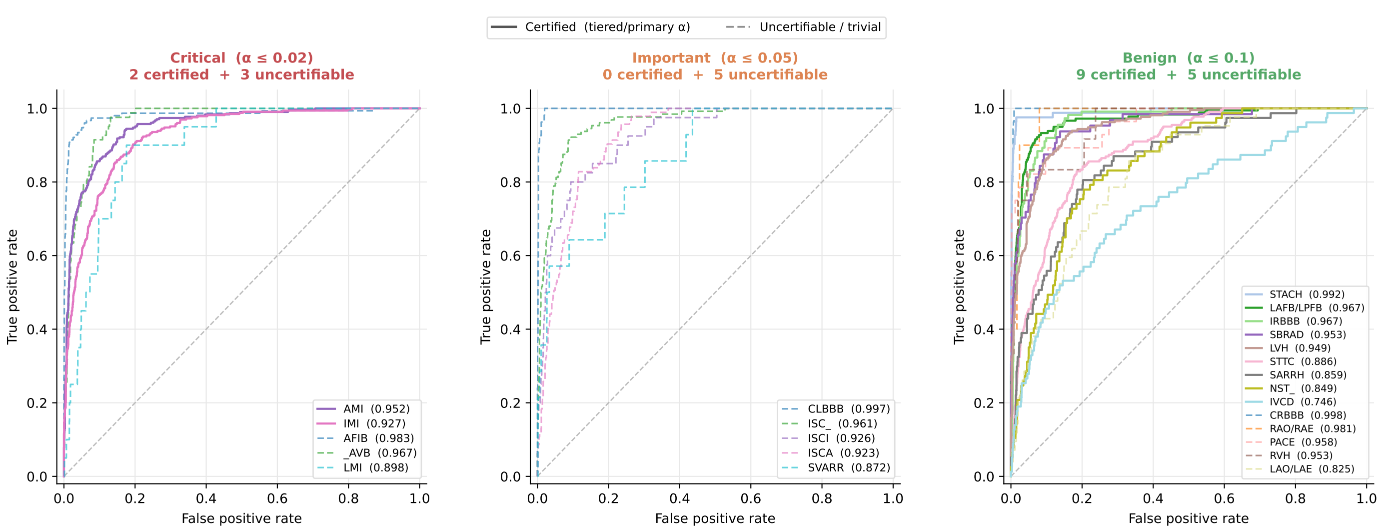

Figure 2: Caption

Table 4
Primary calibration results across clinical tiers. Dashes (—) indicate uncertifiable classes where no valid threshold could be found given the target risk and sample size.

Class	Tier	α	n_{val}^{+}	λ_c	Test FNR	Flag %	Status
AFIB	Critical	0.02	151	—	—	—	uncertifiable
AMI	Critical	0.02	306	0.0060	0.0131	54.0%	certified
IMI	Critical	0.02	326	0.0020	0.0061	76.1%	certified
LMI	Critical	0.02	20	—	—	—	uncertifiable
_AVB	Critical	0.02	83	—	—	—	uncertifiable
CLBBB	Important	0.05	54	—	—	—	uncertifiable
ISCA	Important	0.05	92	—	—	—	uncertifiable
ISCI	Important	0.05	39	—	—	—	uncertifiable
ISC_	Important	0.05	125	—	—	—	trivial ($\lambda = 0$)
SVARR	Important	0.05	15	—	—	—	uncertifiable
CRBBB	Benign	0.10	55	—	—	—	trivial ($\lambda = 0$)
IRBBB	Benign	0.10	112	0.0160	0.0089	26.3%	certified
IVCD	Benign	0.10	78	0.0010	0.0127	96.1%	certified
LAFB/LPFB	Benign	0.10	181	0.0731	0.0615	17.5%	certified
LAO/LAE	Benign	0.10	43	—	—	—	uncertifiable
LVH	Benign	0.10	210	0.0260	0.0514	29.9%	certified
NST_	Benign	0.10	75	0.0010	0.0000	70.9%	certified
PACE	Benign	0.10	29	—	—	—	uncertifiable
RAO/RAE	Benign	0.10	10	—	—	—	uncertifiable
RVH	Benign	0.10	12	—	—	—	uncertifiable
SARRH	Benign	0.10	77	0.0070	0.0260	66.6%	certified
SBRAD	Benign	0.10	64	0.0010	0.0156	51.3%	certified
STACH	Benign	0.10	83	0.0030	0.0244	9.8%	certified
STTC	Benign	0.10	225	0.0310	0.0901	42.4%	certified

Table 5
Failure-Mode Scorecard: Impact of uniform α configurations vs. acuity-weighted tiering. The two failure modes are disjoint, demonstrating that no single scalar α can resolve both simultaneously.

Panel A: Safety failures (false security) [6 classes]

Class	Tier	α_t	Tiered λ_c	Tiered Status	Worst Unif. λ_c	Unif. α	Unif. Status
AFIB	Critical	0.02	—	uncert.	0.0541	0.10	cert.
AMI	Critical	0.02	0.0060	cert.	0.0881	0.10	cert.
IMI	Critical	0.02	0.0020	cert.	0.0671	0.10	cert.
_AVB	Critical	0.02	—	uncert.	0.0020	0.10	cert.
ISCA	Important	0.05	—	uncert.	0.0020	0.10	cert.
ISC_	Important	0.05	—	trivial	0.0070	0.10	cert.

Panel B: Efficiency losses (needlessly weak) [9 classes]

Class	Tier	n^{+}_{val}	Tiered λ_c	Tiered FNR	Unif. 0.02	Unif. 0.05	Unif. 0.10
IRBBB	Benign	112	0.0160	0.0089	uncert.	trivial	cert.
IVCD	Benign	78	0.0010	0.0127	uncert.	uncert.	cert.
LAFB/LPFB	Benign	181	0.0731	0.0615	uncert.	trivial	cert.
LVH	Benign	210	0.0260	0.0514	uncert.	cert.	cert.
NST_	Benign	75	0.0010	0.0000	uncert.	uncert.	cert.
SARRH	Benign	77	0.0070	0.0260	uncert.	uncert.	cert.
SBRAD	Benign	64	0.0010	0.0156	uncert.	uncert.	cert.
STACH	Benign	83	0.0030	0.0244	uncert.	uncert.	cert.
STTC	Benign	225	0.0310	0.0901	uncert.	cert.	cert.

Panel C: Unaffected [9 classes]

Class	Tier	n^{+}_{val}	Tiered Status	Unif. 0.02	Unif. 0.05	Unif. 0.10
LMI	Critical	20	uncert.	uncert.	uncert.	uncert.
CLBBB	Important	54	uncert.	uncert.	uncert.	trivial
ISCI	Important	39	uncert.	uncert.	uncert.	uncert.
SVARR	Important	15	uncert.	uncert.	uncert.	uncert.
CRBBB	Benign	55	trivial	uncert.	uncert.	trivial
LAO/LAE	Benign	43	uncert.	uncert.	uncert.	uncert.
PACE	Benign	29	uncert.	uncert.	uncert.	uncert.
RAO/RAE	Benign	10	uncert.	uncert.	uncert.	uncert.
RVH	Benign	12	uncert.	uncert.	uncert.	uncert.

Summary of Effects

Safety failures (Panel A)	6 classes: Uniform α publishes too-weak guarantees.
Efficiency losses (Panel B)	9 classes: Uniform α discards working guarantees.
Unaffected (Panel C)	9 classes: Outcome identical across all configurations.
Total Affected	**15 / 24 (63%) classes are worse off under uniformity.**

Table 6
Performance Comparison of Conformal Calibration Strategies

Method	Violation Rate	Mean λ_c	Mean FNR	Status
Hoeffding + grid-Bonferroni	0.000	0.4178	0.0327	PASS
Hoeffding + monotone boundary	0.000	0.4734	0.0667	PASS
Binomial + monotone boundary (*Adopted*)	**0.078**	**0.4974**	**0.0876**	**PASS**

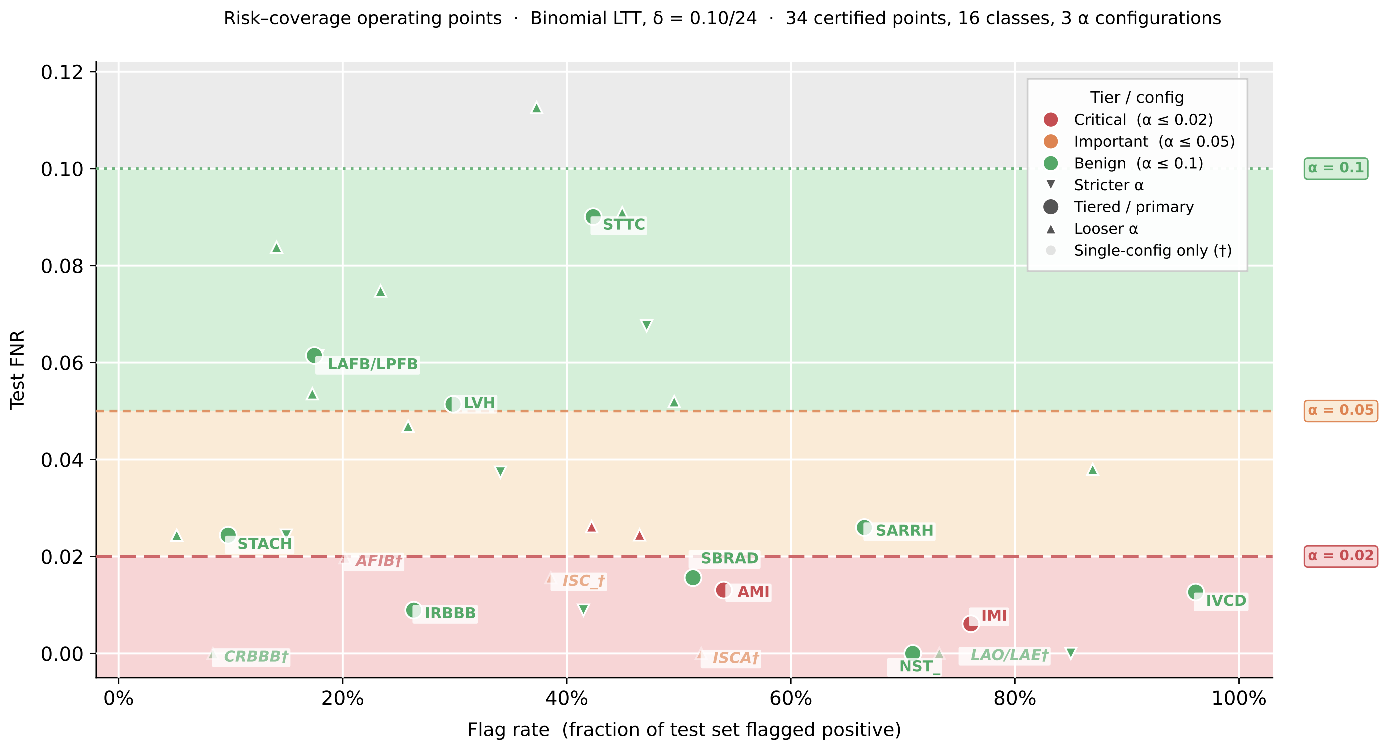

Figure 3: Caption

targets are therefore structurally required rather than a tuning preference. The disjointness of the efficiency-loss and safety-failure sets (Table 5) is consistent with this hypothesis: the two failure modes occur at opposite ends of the acuity range and do not share any class, so no scalar α can lie above the Benign-tier classes that require a loose target while simultaneously lying below the Critical- and Important-tier classes that require a strict one. A uniform $\alpha = 0.10$ preserves Benign-tier certifiability at the cost of issuing 6 guarantees too weak for their clinical tier, whereas a uniform $\alpha = 0.02$ removes those safety failures but discards 9 otherwise-valid Benign-tier guarantees. The tiered assignment is the only configuration examined that incurs neither failure mode.

The certified Benign-tier results constitute the operationally usable output of the procedure: 9 classes with test FNRs below the 0.10 target and flag rates spanning an order of magnitude, indicating that selectivity (the fraction of records referred for review) varies substantially across classes even at a common target. The two certified Critical-tier classes (AMI, IMI) achieved their stricter target only at low thresholds, with correspondingly high flag rates (54.0%, 76.1%); the guarantee for these classes is therefore tight but minimally selective. The juxtaposition of high AUROC with non-certifiability (AFIB, CLBBB) indicates that certifiability under this framework is governed primarily by the number of positive calibration examples available to the binomial test, not by the discriminative quality of the underlying classifier — a class can separate well yet lack the calibration-fold support needed to reject the null at the corrected error rate.

This sample-size dependence is the principal limitation of the present design. Certification draws on a single patient-stratified fold (fold 9, n = 2,183) that also serves as the training-time validation set, and the rarest high-acuity classes are precisely those for which this fold supplies too few positive examples to certify at a strict target (AFIB, _AVB, and the entire Important tier). The framework consequently certifies most reliably for common, lower-acuity findings and least reliably for the rare, higher-acuity findings where a guarantee would carry the greatest clinical value. A secondary limitation is the reuse of the validation fold as the calibration fold, which narrows the exchangeability assumption underlying the conformal guarantee, as noted in Section 4.1.

A concrete direction for future work follows directly from this limitation: augmenting the calibration set for rare high-acuity classes with exchangeably-pooled positive examples from external 12-lead datasets (e.g., CPSC2018, Chapman–Shaoxing, Georgia12EC), with coverage evaluated on a test population drawn from the same pooled sources to preserve exchangeability. Such pooling would raise the positive-count available to the binomial test for classes such as AFIB and _AVB above the certification floor, allowing a direct test of whether their non-certifiability in the present study reflects a dataset-scale constraint, as the within-PTB-XL evidence suggests, rather than a limitation of the method itself.

7. Conclusion

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